

Fairness Audits of Institutional Risk Models in Deployed ML Pipelines

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Abstract

Fairness audits of institutional risk models are critical for understanding how deployed machine learning pipelines allocate resources. Drawing on multi-year collaboration with a Canadian college, we present a replica-based audit of a deployed Early Warning System (EWS), replicating its model using institutional training data and design specifications. We evaluate disparities by gender, age, and residency status across the full pipeline (including training data, model predictions, and post-processing) using standard fairness metrics.

Our audit reveals systematic misallocation: younger, male, and international students are disproportionately flagged for support, even when many ultimately succeed, while older and female students with comparable dropout risk are under-identified. Post-processing amplifies these disparities by collapsing heterogeneous probabilities into percentile-based risk tiers.

This work provides a replicable methodology for auditing institutional ML systems. We show how disparities emerge and compound across stages, highlighting the importance of evaluating construct validity alongside statistical fairness. Our findings underscore the need for audits of in-use institutional ML systems and motivate future directions including intersectional analyses and post-hoc fairness interventions.

Keywords: Fairness auditing, educational AI, algorithmic bias, machine learning, higher education, early warning systems

1. Introduction

Algorithmic systems designed to predict individual risk have become increasingly common across public sector domains, including child welfare [1], criminal justice [2], and homelessness [3]. Promising efficiency and scalability [4], these systems are often adopted under resource scarcity. Yet concerns about bias and algorithmic harm persist. While existing research has explored sociotechnical contexts of algorithmic deployment, relatively few studies have audited deployed systems already influencing public decisions. This gap is particularly notable in educational contexts, where access to real institutional data, design documentation, and operational systems remains limited [5].

In this paper, we present a fairness audit of an Early Warning System (EWS) used in a Canadian public higher education institution to predict student success. While prior research has examined perceptions of fairness and trust in AI [6], little attention has been paid to how these systems function internally—how they encode and reproduce definitions of risk, and with what consequences. Drawing on multi-year collaboration with a Canadian college, we replicate and audit a deployed EWS trained on real student data. Our analysis focuses on how model design choices and operational definitions of risk produce disparities across gender, age, and residency status.

Early Warning Systems predict students' likelihood of success to triage them into intervention categories aimed at improving retention [7]. Our previous research found that such

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systems are often introduced in response to policy pressures and resource constraints, affecting both advising practice and support distribution [8]. Despite growing interest in how predictive systems may amplify inequities, little is known about how "risk" is operationalized or how such operationalization translates into resource allocation.

In this study, we ask:

- **RQ1:** How are different population groups represented and treated by the Early Warning System's (a) training data, (b) predictions, and (c) interpretation?
- **RQ2:** How does the Early Warning System operationalize risk in terms of student drop-out?

To answer these questions, we replicated the institution's EWS using its training data and documented design specifications, then audited the system across the pipeline, focusing on disparities across sensitive groups.

The major contributions of this paper include:

- A generalizable **framework for fairness audits of institutional risk models**, showing how disparities can be traced across the full ML pipeline.
- One of the first **empirical audits of a deployed educational Early Warning System**, replicating the institutional model with real student intake data and documented design specifications.
- Quantification of disparities in accuracy, false positive/negative rates, and fairness metrics across gender, age, and residency groups.
- Analysis of the **construct validity** of dropout as a proxy for "student need," demonstrating how target definition interacts with fairness metrics.
- Design implications for **trustworthy ML auditing practice**, including construct validity evaluation, intersectional analysis, and post-hoc fairness interventions.

Together, these contributions advance fairness audits of real-world ML systems, offering a replicable case study of how fairness metrics, construct validity, and institutional context intersect in EWS deployment.

2. Related Work

2.1. Fairness audits of algorithmic risk prediction systems

Our work builds on extensive research examining algorithmic systems in child welfare [1], criminal justice [2], and homelessness [3]. This literature has documented how predictive systems can perpetuate disparities through problem formulation and target variable definition. Scholars emphasize the importance of transparency in design choices and incorporating community feedback throughout development.

Methodologically, academic audits often face constraints due to proprietary models or limited access, leading researchers to rely on in-house replications or component-level analyses. Raji et al. argue that internal algorithmic audits are essential for accountability across the AI lifecycle [9], while Selbst et al. emphasize that audits must account for sociotechnical context since bias often arises from interactions between algorithms and institutions [10]. Our replication-based approach builds on this tradition by combining fidelity to original design specifications with systematic fairness evaluation. Foundational results show that fairness criteria are mutually incompatible in general [11], motivating auditing frameworks that extend beyond outputs to interrogate data, target definitions, and institutional choices [12].

2.2. Target variable choice and construct validity

A critical aspect of fairness is how target variable choice shapes both predictive performance and equity outcomes. Obermeyer et al.'s influential study of healthcare allocation

algorithms demonstrated that differential mismeasurement across groups can arise from poor proxy choices—using healthcare costs rather than actual health needs led to systematic underestimation of Black patients’ care requirements [13]. Passi and Barocas extend this analysis, showing how problem formulation decisions embed values and priorities that compound fairness concerns [14]. Jacobs and Wallach’s work on measurement modeling highlights how institutional convenience often takes precedence over construct validity in ML system design [15]. Our analysis applies these frameworks to educational risk prediction, examining how defining "risk" as program non-completion shapes fairness outcomes.

2.3. Early Warning Systems in higher education

Within higher education, Early Warning Systems predict student success to guide interventions and improve retention. Perdomo et al.’s evaluation of Wisconsin’s system provides a particularly relevant parallel, demonstrating how prediction-based approaches may fail to achieve stated goals due to problem formulation issues [7]. Our prior ethnographic work at this site documented how EWS adoption automated advising decisions and affected resource allocation [8]. Despite growing deployment, relatively few studies have combined technical fairness audits with access to real institutional data, design documentation, and operational context. Our work contributes to this emerging area by demonstrating how replica-based audits can trace disparities across the complete ML pipeline in educational settings.

3. Methods

Our research site is a public college in Canada where an Early Warning System (EWS) was deployed to predict student retention. Through multi-year institutional collaboration, we obtained training data and design specifications from our prior ethnographic work [8], enabling replica-based audit of the deployed system. This methodology provides three advantages: access to actual training data reveals distributional patterns obscured in public datasets; documented institutional decisions trace how organizational priorities embed in technical specifications; operational context enables assessing fairness as resource allocation affecting real students.

3.1. Data and Modeling

After research ethics approval, we obtained student data spanning 2011–2019, resulting in 102,353 records (61,375 domestic, 40,978 international students). To replicate the intake EWS, we filtered to first-semester students with features available at intake. Features differed for international (English test scores) and domestic (high school grades) students due to distinct admissions processes. Following institutional design, we trained separate XGBoost models for each population.

We retained 45 features covering intake, program, and admissions attributes after removing those with substantial missing data. We acknowledge this decision has limitations: missingness can be informative, and if correlated with sensitive attributes, removal could introduce bias. Sensitive attributes (age, gender, residency, funding, first-generation status) were included as model features in line with replicated system. Race/ethnicity and disability data were unavailable. The binary target labeled students *Successful* if they completed their program within the allowable period, and *Unsuccessful* otherwise (withdrawal, failure, transfer). Data was split chronologically (70% training, 15% validation, 15% test), balanced using SMOTE [16], and XGBoost classifiers were trained using grid search.

The EWS pipeline consists of three stages: (1) intake data collection, (2) separate XGBoost models producing success probabilities, and (3) percentile-based categorization into Low Risk (top 50%), Medium Risk (next 27%), and High Risk (bottom 23%) tiers. We

trained separate models for domestic versus international students following institutional practice, as these populations have distinct admissions processes and feature availability. We acknowledge that finer-grained models might reveal different patterns. Test predictions were sorted by probability and assigned to risk categories, yielding 15,461 students (9,209 domestic, 6,252 international) for fairness analysis.

3.2. Fairness Metrics

We evaluate disparities using standard fairness metrics: Statistical Parity Difference (SPD), Equal Opportunity Difference (EOD), False Positive Rate gaps (ΔFPR), and Calibration Error (CE). For each sensitive attribute, we compute pairwise disparities and report maximum absolute differences, following prior audit practice [9, 17].

3.3. Operationalizing Risk

A central methodological decision is the target outcome definition. The institution’s EWS was designed to predict whether a student would be *unsuccessful*, defined as failure to complete the credential originally enrolled in. This binary outcome collapses dropout, transfer, and program change into a single label.

This operationalization treats dropout as a proxy for the construct of *student risk* or *need for support*. While measurable, this proxy has limitations: not all students who drop out require support, and not all students who persist are free of need. Following Jacobs and Wallach [15], we treat this as a construct validity issue. For this audit, we adopt the institution’s definition to faithfully replicate the deployed system and evaluate fairness under the same operational criteria. We return to the implications of this proxy in Section 5.

4. Findings

We conduct a systematic fairness audit across the EWS pipeline, evaluating disparities in data representation, model predictions, and post-processing interpretation. Our analysis focuses on three demographic attributes: gender, age, and residency status.

4.1. Baseline Data Disparities

Training data reveals significant group-level differences in program completion rates. International students achieve 85% success compared to 67% for domestic students ($\chi^2 = 2847.3$, $p < 0.001$). Female students significantly outperform male students in both populations: domestic (73% vs. 59%, $\chi^2 = 1205.4$, $p < 0.001$) and international (89% vs. 82%, $\chi^2 = 142.7$, $p < 0.001$). Age effects follow consistent patterns, with students 26+ achieving higher completion rates than those ≤ 20 years across both populations.

These baseline disparities establish the foundation for subsequent model learning and bias amplification documented in the following sections.

4.2. Model Performance and Fairness Metrics

4.2.1. Overall Performance Disparities

International student models achieve significantly higher accuracy (91%) than domestic models (82%, $p < 0.001$). This performance gap translates to differential access to intervention resources, as unsuccessful international students are $1.08\times$ more likely to be correctly identified. Both models rely heavily on program-level features (credential type, program length) rather than individual student characteristics, raising questions about whether student-level risk prediction is necessary when program-level patterns dominate.

4.2.2. Gender-Based Disparities

Table 1. Model performance by gender.

Gender	Domestic Students				International Students			
	Accuracy	FPR	FNR	F1	Accuracy	FPR	FNR	F1
Female	0.84	0.32	0.10	0.89	0.92	0.26	0.05	0.96
Male	0.81	0.23	0.17	0.83	0.91	0.18	0.08	0.94

Female students receive systematically different treatment: higher false positive rates (domestic: 32% vs 23%, international: 26% vs 18%) but lower false negative rates, creating differential intervention allocation patterns (Table 1). All metrics exceed commonly used fairness thresholds ($SPD > 0.1$), indicating substantial gender-based disparities.

4.2.3. Age-Based Disparities

Students aged 36-40 show false positive rates above 0.60, while younger students aged 19-20 exhibit false negative rates of 0.21 (Table 2), indicating systematic age-based disparities in prediction errors.

4.3. Post-Processing and Risk Categorization Effects

The EWS converts probabilities into three risk tiers, amplifying biases present in model predictions.

4.3.1. Threshold Effects and Calibration

Probability thresholds (0.80 for Medium Risk, 0.39 for High Risk) mean students with vastly different success probabilities receive identical interventions. The Medium Risk category exhibits poor calibration (Brier Score: 0.18 vs. 0.04 for Low Risk), with approximately 21% of High Risk students ultimately succeeding, producing systematic resource misallocation.

4.3.2. Amplified Group Disparities

Post-processing amplifies demographic disparities beyond those observed in raw model predictions:

Gender Effects: Unsuccessful male students are 10% more likely to be categorized as High Risk than females (74% vs. 63%, $p = 1.56 \times 10^{-13}$), corresponding to roughly 300 additional male students per year receiving intensive interventions. This illustrates how categorical thresholds magnify gender disparities.

Table 2. Error rates by age group.

Age Group	Domestic Students			International Students			Differences	
	FPR	FNR	F1	FPR	FNR	F1	ΔFPR	ΔFNR
0-18	0.21	0.18	0.84	0.19	0.09	0.93	0.02	0.09
19-20	0.20	0.21	0.81	0.18	0.10	0.92	0.02	0.11
21-25	0.29	0.10	0.88	0.18	0.06	0.96	0.11	0.05
26-30	0.33	0.09	0.90	0.25	0.03	0.97	0.09	0.06
31-35	0.41	0.07	0.92	0.29	0.05	0.96	0.12	0.02
36-40	0.61	0.08	0.90	0.70	0.02	0.97	-0.09	0.05
41-50	0.53	0.05	0.91	0.69	0.02	0.95	-0.16	0.25

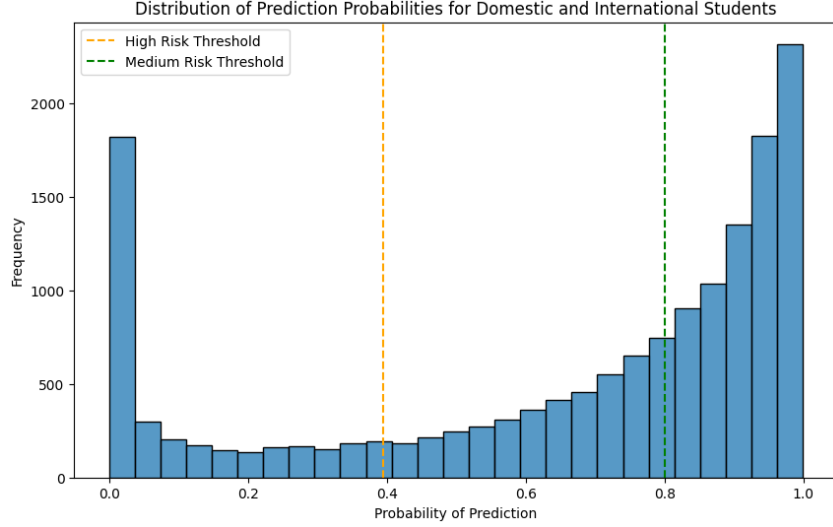


Figure 1. Distribution of success prediction probabilities from the EWS, showing bi-modal clustering. The orange dashed line (≈ 0.4) marks the High Risk threshold and green dashed line (≈ 0.8) marks the Medium Risk threshold.

Age Effects: Students ≤ 25 years have significantly higher probability of High Risk categorization than those 36+ (94% vs. 75%). Older students aged 36-40 show false positive rates above 0.60, indicating many who ultimately succeed receive high-intensity interventions.

Residency Effects: Unsuccessful international students are $1.12\times$ more likely to receive High Risk classification than domestic students (76% vs 68%), exceeding the $1.08\times$ difference in raw model accuracy. This amplification indicates post-processing accentuates group disparities.

4.3.3. Intersectional Analysis

Intersectional effects compound disparities. Female international students aged 41-45 have only 14% probability of High Risk classification, compared to 100% for female domestic students aged 36-40 with similar outcomes, illustrating how overlapping identities produce compounding harms invisible in single-attribute analyses.

4.4. Construct Validity of Risk Operationalization

The EWS operationalizes "risk" as the probability of program non-completion, conflating dropout, transfer, and program change into a single unsuccessful outcome. This proxy measure exhibits systematic measurement error: not all students who leave programs require intervention support, while some who persist may still need substantial assistance. The percentile-based categorization further weakens construct validity by treating prediction uncertainty (Medium Risk category) as equivalent to moderate intervention need, effectively institutionalizing model error as a resource allocation rule.

5. Discussion

Our audit demonstrates how disparities arise and compound across the full machine learning pipeline. While prior work has identified fairness challenges in educational AI

broadly, our findings provide an empirical case study of how these challenges manifest in practice. By examining training data distributions, model outputs, and percentile-based risk categories, we show that residency, gender, and age groups experience systematically different error rates and intervention allocations.

5.1. Fairness Across the EWS Pipeline (RQ1)

Group-level disparities are structurally embedded across the EWS pipeline. Base success rate differences are learned by models and amplified by percentile-based interpretation, producing systematic over-allocation to younger, male, and international students and under-allocation to older, female, and domestic students. This highlights a core trade-off: the EWS emphasizes group-level allocation reflecting demographic trends rather than individual-level need. Percentile thresholds assign identical labels to students with different probabilities (e.g., 0.45 vs. 0.75), eliminating sensitivity to individual variation [7, 11]. Recent research on hybrid fairness-aware approaches [18] suggests group-conditional calibration or multi-objective optimization could better balance group and individual fairness in EWS contexts.

5.2. Operationalizing Risk: Three Conceptual Issues

Beyond measurable disparities, our audit reveals three distinct but related problems with how "risk" is defined and operationalized in this EWS. Each compounds the fairness issues documented in RQ1.

5.2.1. Task Formulation Mismatch

The EWS predicts dropout probability, but practitioners interpret these predictions as indicators of who would benefit most from support. These are distinct quantities. Students may drop out for reasons unrelated to academic need—transfer, financial constraints, or personal circumstances. Conversely, students who persist may still struggle significantly. This formulation error creates systematic measurement bias: the model optimizes for predicting an observable outcome (dropout) rather than the unobservable construct of interest (intervention benefit). Our findings show this mismatch interacts with fairness—groups with different baseline completion patterns are disproportionately labeled "at risk" even when prediction confidence suggests more nuanced variation.

5.2.2. Institutional Priorities versus Student-Centered Success

The choice of program non-completion as the target variable reflects institutional priorities, maintaining enrollment and revenue, rather than student-defined success or wellbeing. As our prior ethnographic work documented, the EWS was implemented during funding cuts and tuition freezes, creating pressure to improve retention metrics. However, students and advisors may understand "success" differently: acquiring transferable skills, finding appropriate career paths, maintaining mental health, or achieving personal goals beyond credential completion. A student who transfers appears as "unsuccessful" in the model, yet may be progressing toward their actual goals. We recognize that measuring alternative outcomes (wellbeing, skill acquisition) poses significant practical and financial challenges. However, these constraints should inform system design decisions such as acknowledging measurement limitations or involving stakeholders in defining proxies, rather than justifying optimizing for available but flawed proxies.

5.2.3. Prediction Uncertainty Misinterpreted as Moderate Need

The EWS converts continuous probabilities into three risk tiers using percentile thresholds. Students with success probabilities ranging from 0.41 to 0.80 are all assigned "Medium Risk" and receive identical intervention levels. Our findings show this category exhibits the poorest calibration (Brier Score: 0.18 vs. 0.04 for Low Risk). In effect, the system institutionalizes prediction uncertainty as a resource allocation rule: practitioners interpret Medium Risk scores as indicating moderate support needs, when in reality these scores reflect the model's lack of confidence. This amplifies demographic disparities—percentile-based categorization increases the gap in High Risk classification between male and female students from 36% (raw predictions) to 40% (post-processed categories).

These three issues—task formulation, values misalignment, and uncertainty interpretation—are analytically distinct but practically interconnected. Together, they demonstrate that fairness in EWS cannot be achieved through technical fixes alone; it requires reconsidering the prediction task, whose priorities are embedded in outcome definitions, and how predictions are translated into resource allocation decisions.

6. Implications and Limitations

6.1. Implications for Educational AI Systems

Our audit demonstrates how algorithmic risk definitions systematically shape intervention allocation. The model produces binary predictions, yet academic success is multidimensional. When allocation patterns favor some groups (Section 5.1), they communicate differential institutional value.

A fundamental misalignment exists: *who needs help* differs from *who is likely to drop out*. The EWS assigns intensive interventions to students most confidently predicted to fail, while those with intermediate probabilities receive less support despite high calibration error, resulting in under-allocation to female and older students.

Program-level features dominate predictions, suggesting the system optimizes for institutional retention rather than individual need. Implemented during funding cuts, the EWS reflects policy pressures around financial sustainability. These findings necessitate participatory design engaging advisors, students, and policymakers to align models with educational goals.

6.2. Implications for Trustworthy ML

Our study demonstrates three critical insights for trustworthy ML systems. First, fairness audits must examine the full ML pipeline, as disparities are introduced at the data stage, learned by models, and amplified during post-processing. Second, construct validity is central to fairness: when predictive targets are imperfect proxies, fairness metrics alone provide incomplete equity assessment. Third, replica-based audits of institutional risk models provide actionable insights for public-sector deployments where ML pipelines allocate scarce resources. Trustworthy ML requires not only technical improvements but also systematic auditing practices integrating institutional goals, construct validity, and post-deployment evaluation.

6.3. Limitations

Our analysis is based on a *replica* trained on the same data and design protocol, not the original production model. While this provides strong fidelity, results may diverge in unknown ways. We lack data on actual intervention delivery or student responses, so our audit evaluates predictions and categorizations, not downstream effects. The outcome label

conflates dropout, transfer, and program change, raising construct validity concerns. Additionally, our replica-based methodology addresses calls for grounded socio-technical analyses but faces constraints of partial system access common in academic audits of proprietary institutional systems.

7. Conclusion

This paper demonstrates a replication-based methodology for auditing deployed educational ML systems using real institutional training data and documented design specifications. Our analysis quantifies fairness disparities across the complete EWS pipeline: demographic differences in training data (18 percentage point gap in completion rates between female and male domestic students), systematic prediction errors (false positive rates varying by 40 percentage points across age groups), and post-processing amplification (10% increase in High Risk classification differences between gender groups).

These findings contribute three methodological advances for educational ML auditing: (1) demonstration that institutional collaboration enables measurement of bias amplification across technical pipeline stages, (2) quantification of how percentile-based thresholds systematically compound prediction disparities, and (3) empirical evidence that construct validity failures interact with demographic bias to produce inequitable resource allocation.

For the trustworthy ML community, our findings demonstrate that fairness audits must extend beyond standard statistical metrics to include construct validity evaluation and an understanding of how post-processing decisions amplify disparities. By framing our analysis as an audit of institutional risk models, we provide a replicable methodology for examining ML pipelines that allocate scarce resources in education and other public-sector domains. This work highlights how auditing deployed systems can both surface disparities and guide the design of more equitable, trustworthy ML models in practice.

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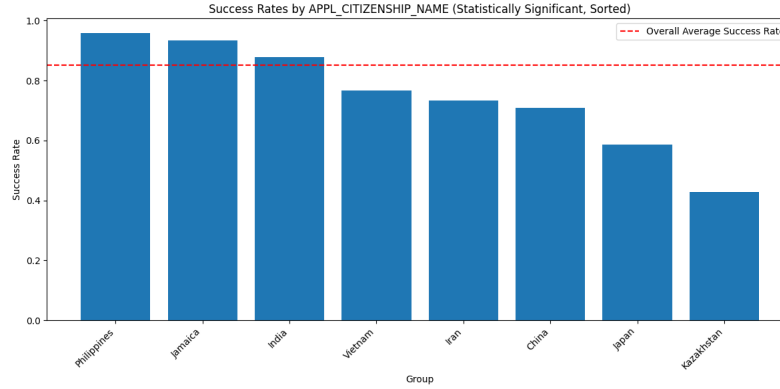


Figure 2. Success Rates by Citizenship

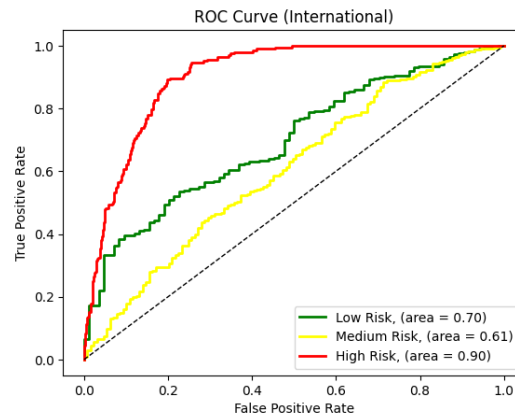


Figure 3. International ROC Curve

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Appendix A. Tables and Charts from Findings

Age Groups	SPD	EOD	AOD	DI
21 vs 36	-0.1710	-0.1660	-0.0855	0.8034
21 vs 31	-0.1286	-0.1447	-0.0643	0.8446
21 vs 41	-0.1665	-0.1551	-0.0833	0.8076
21 vs 0	0.1143	0.0997	0.0571	1.1955
21 vs 51	-0.1095	-0.1138	-0.0548	0.8645
21 vs 26	-0.0607	-0.0681	-0.0303	0.9201
21 vs 19	0.1669	0.1589	0.0834	1.3137
21 vs 66	-0.3012	-0.3971	-0.1506	0.6988
21 vs 61	-0.1194	-0.0335	-0.0597	0.8541
36 vs 31	0.0424	0.0214	0.0212	1.0512
36 vs 41	0.0045	0.0109	0.0022	1.0051
36 vs 0	0.2852	0.2657	0.1426	1.4880
36 vs 51	0.0614	0.0522	0.0307	1.0760
36 vs 26	0.1103	0.0979	0.0552	1.1453
36 vs 19	0.3378	0.3249	0.1689	1.6351
36 vs 66	-0.1303	-0.2311	-0.0651	0.8697
36 vs 61	0.0516	0.1325	0.0258	1.0630
31 vs 41	-0.0379	-0.0104	-0.0190	0.9562
31 vs 0	0.2428	0.2444	0.1214	1.4155
31 vs 51	0.0190	0.0309	0.0095	1.0235
31 vs 26	0.0679	0.0766	0.0340	1.0894
31 vs 19	0.2954	0.3035	0.1477	1.5554
31 vs 66	-0.1726	-0.2524	-0.0863	0.8274
31 vs 61	0.0092	0.1112	0.0046	1.0112
41 vs 0	0.2808	0.2548	0.1404	1.4804
41 vs 51	0.0570	0.0413	0.0285	1.0705
41 vs 26	0.1059	0.0870	0.0529	1.1394
41 vs 19	0.3334	0.3140	0.1667	1.6268
41 vs 66	-0.1347	-0.2420	-0.0674	0.8653
41 vs 61	0.0471	0.1216	0.0236	1.0576
0 vs 51	-0.2238	-0.2135	-0.1119	0.7231
0 vs 26	-0.1749	-0.1678	-0.0875	0.7697
0 vs 19	0.0526	0.0592	0.0263	1.0989
0 vs 66	-0.4155	-0.4968	-0.2077	0.5845
0 vs 61	-0.2337	-0.1332	-0.1168	0.7144
51 vs 26	0.0489	0.0457	0.0244	1.0644
51 vs 19	0.2764	0.2727	0.1382	1.5197
51 vs 66	-0.1917	-0.2833	-0.0958	0.8083
51 vs 61	-0.0098	0.0803	-0.0049	0.9880
26 vs 19	0.2275	0.2270	0.1138	1.4277
26 vs 66	-0.2406	-0.3290	-0.1203	0.7594
26 vs 61	-0.0587	0.0346	-0.0294	0.9282
19 vs 66	-0.4681	-0.5560	-0.2340	0.5319
19 vs 61	-0.2863	-0.1924	-0.1431	0.6501
66 vs 61	0.1818	0.3636	0.0909	1.2222

Table 3. Domestic Fairness Metrics Across Age Groups

Age Groups	SPD	EOD	AOD	DI
26 vs 21	0.0475	0.0453	0.0238	1.0556
26 vs 36	-0.0575	-0.0402	-0.0287	0.9401
26 vs 31	0.0073	0.0076	0.0036	1.0081
26 vs 19	0.1844	0.2069	0.0922	1.2568
26 vs 41	-0.0466	-0.0039	-0.0233	0.9509
26 vs 0	0.1226	0.1346	0.0613	1.1572
26 vs 51	-0.0977	-0.0374	-0.0489	0.9023
21 vs 36	-0.1050	-0.0855	-0.0525	0.8906
21 vs 31	-0.0403	-0.0377	-0.0201	0.9550
21 vs 19	0.1369	0.1616	0.0684	1.1906
21 vs 41	-0.0941	-0.0492	-0.0471	0.9008
21 vs 0	0.0751	0.0893	0.0375	1.0963
21 vs 51	-0.1452	-0.0826	-0.0726	0.8548
36 vs 31	0.0647	0.0478	0.0324	1.0723
36 vs 19	0.2419	0.2471	0.1209	1.3369
36 vs 41	0.0109	0.0363	0.0054	1.0115
36 vs 0	0.1800	0.1748	0.0900	1.2309
36 vs 51	-0.0402	0.0029	-0.0201	0.9598
31 vs 19	0.1771	0.1993	0.0886	1.2467
31 vs 41	-0.0539	-0.0115	-0.0269	0.9432
31 vs 0	0.1153	0.1270	0.0577	1.1479
31 vs 51	-0.1050	-0.0449	-0.0525	0.8950
19 vs 41	-0.2310	-0.2108	-0.1155	0.7566
19 vs 0	-0.0618	-0.0723	-0.0309	0.9207
19 vs 51	-0.2821	-0.2442	-0.1410	0.7179
41 vs 0	0.1692	0.1385	0.0846	1.2170
41 vs 51	-0.0511	-0.0335	-0.0255	0.9489
0 vs 51	-0.2203	-0.1719	-0.1101	0.7797

Table 4. International Fairness Metrics Across Age Groups

	Domestic Students	International Students	Difference
High Risk	0.81	0.75	0.06
Medium Risk	0.72	0.81	-0.09
Low Risk	0.92	0.98	-0.06

Table 5. Accuracy by Risk Category for Domestic and International Students

Risk Category	Domestic Students					International Students				
	FP Count	FN Count	FP Rate	FN Rate	F1 Score	FP Count	FN Count	FP Rate	FN Rate	F1 Score
Low Risk	380	0	1.00	0.00	0.95	86	0	1.00	0.00	0.95
Medium Risk	523	39	0.77	0.13	0.82	186	111	0.79	0.197	0.83
High Risk	0	486	0.00	1.00	0.00	0	237	0.00	1.00	0.00

Table 6. Error Rates for Domestic and International Students by Risk Category

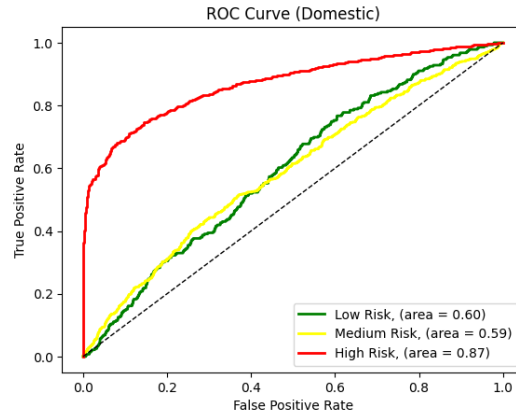


Figure 4. Domestic ROC Curve

Risk Category	Label	Program Type				
		Bachelor Degree	Advanced Diploma	Certificate	Diploma	Graduate Certificate
High Risk	True Negative	0.00	0.22	0.16	0.58	0.03
	False Negative	0.01	0.22	0.16	0.53	0.08
Medium Risk	<i>Combined</i>	<i>0.01</i>	<i>0.17</i>	<i>0.16</i>	<i>0.56</i>	<i>0.10</i>
	True Negative	0.00	0.16	0.21	0.58	0.06
	False Negative	0.01	0.16	0.16	0.59	0.08
	False Positive	0.01	0.20	0.19	0.53	0.07
	True Positive	0.00	0.17	0.15	0.57	0.11
Low Risk	False Positive	0.05	0.11	0.19	0.49	0.15
	True Positive	0.03	0.08	0.16	0.41	0.32

Table 7. Program Credential

Risk Level	Gender	Domestic Accuracy	International Accuracy	Difference
High Risk	F	0.80	0.71	0.08
	M	0.82	0.77	0.05
Medium Risk	F	0.75	0.81	-0.07
	M	0.70	0.81	-0.11
Low Risk	F	0.93	0.98	-0.05
	M	0.90	0.98	-0.08

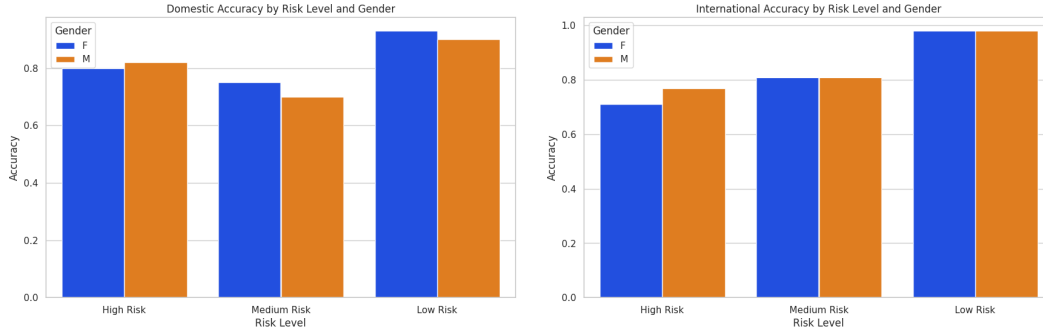


Table 8. Domestic and International Accuracy by Risk Level and Gender

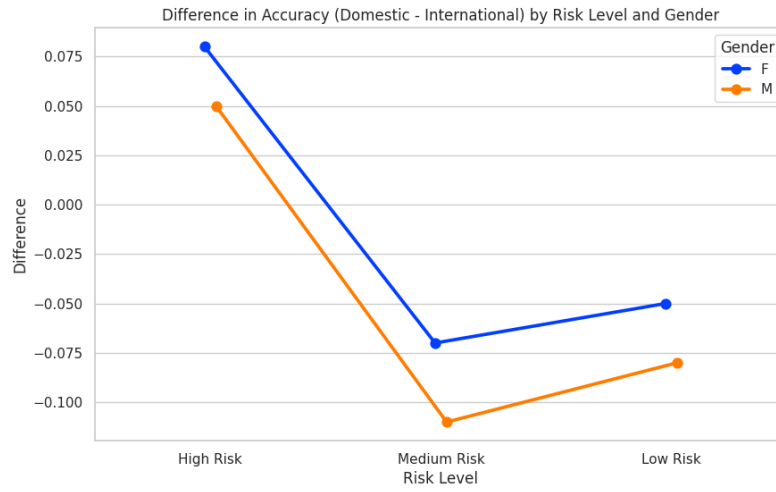


Figure 5. Accuracy Difference by Gender

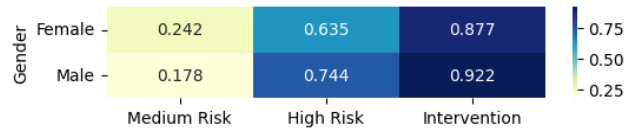


Figure 6. Probability of Risk Categorization by Gender

Gender	Domestic		International		Differences	
	FP Rate	FN Rate	FP Rate	FN Rate	FP	FN
F	0.7727	0.1200	0.7619	0.1005	0.0108	0.0195
M	0.7542	0.1485	0.8056	0.1153	-0.0513	0.0333
N	1.0000	0.0000	0.0000	0.0000	1.0000	0.0000

Table 9. False Positive (FP) and False Negative (FN) Rates by Gender for 'Medium Risk' Domestic and International Students

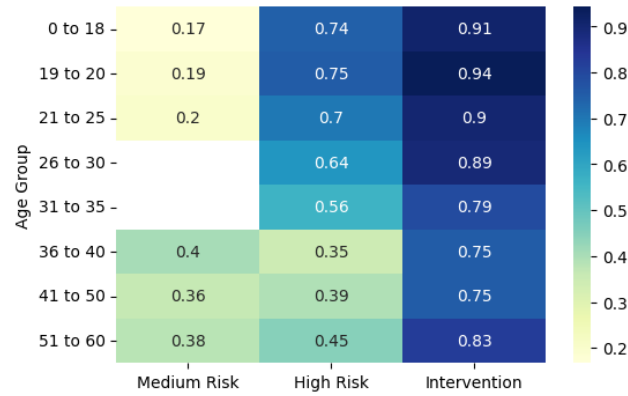


Figure 7. Probability of Risk Categorization by Age Group

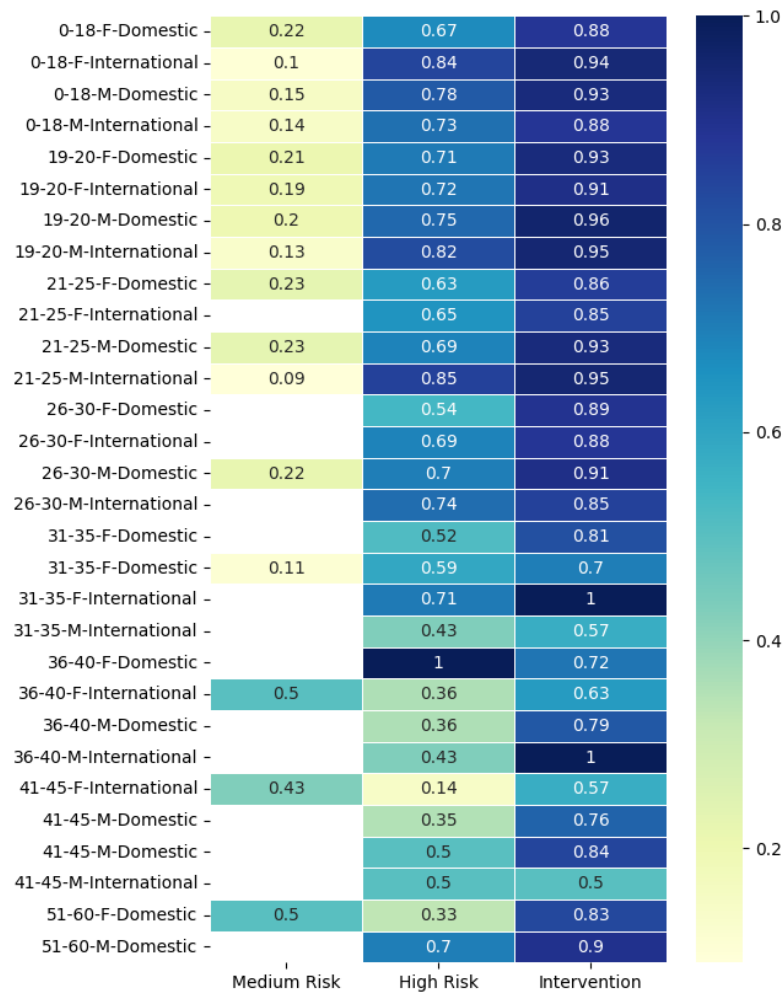


Figure 8. Probability of Risk Categorization by Gender, Age Group, and Status

Age Group	Domestic		International		Differences	
	FP Rate	FN Rate	FP Rate	FN Rate	FP	FN
0-18	0.7118	0.1799	0.7000	0.1093	0.0118	0.0706
19-20	0.6994	0.1507	0.8182	0.1057	-0.1188	0.0450
21-25	0.8218	0.1159	0.7647	0.1115	0.0571	0.0045
26-30	0.8169	0.1128	0.7500	0.1310	0.0669	-0.0182
31-35	0.7778	0.0800	0.7143	0.1136	0.0635	-0.0336
36-40	0.9063	0.1356	1.0000	0.0769	-0.0938	0.0587
41-50	0.8235	0.0522	1.0000	0.0000	-0.1765	0.0522
51-60	0.6364	0.0870	-	-	-	-
61+	0.6667	0.0000	-	-	-	-

Table 10. False Positive (FP) and False Negative (FN) Rates by Age Group for 'Medium Risk' Domestic and International Students

Status in Canada	FP Rate	FN Rate
Canadian Citizen	0.7686	0.1347
Conv Refugee/Protected person/Work Permit/Diplomat	0.6250	0.1667
Native Ancestry (Indian, Metis, Inuit)	1.0000	0.0000
Permanent Resident or Landed Immigrant	0.7292	0.1108
Study Permit (student visa)	0.7820	0.1099

Table 11. FP and FN Rates with Differences by Status in Canada for Domestic and International Models

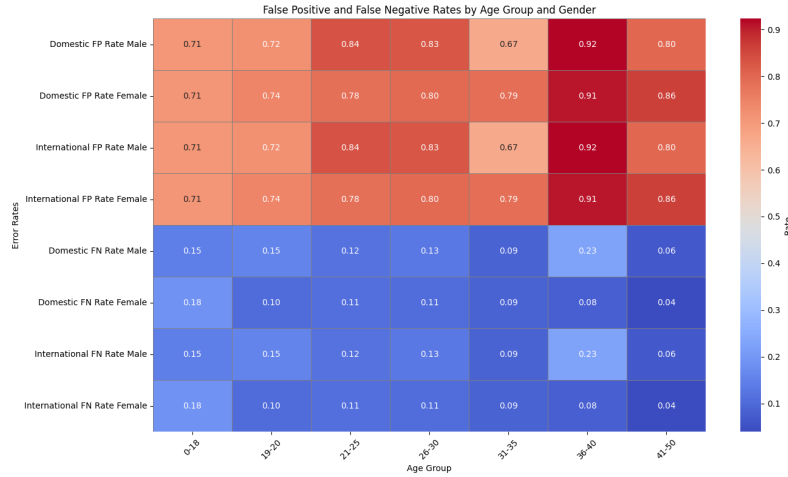


Figure 9. False Positive and False Negative Rates by Age Group and Gender, Medium Risk

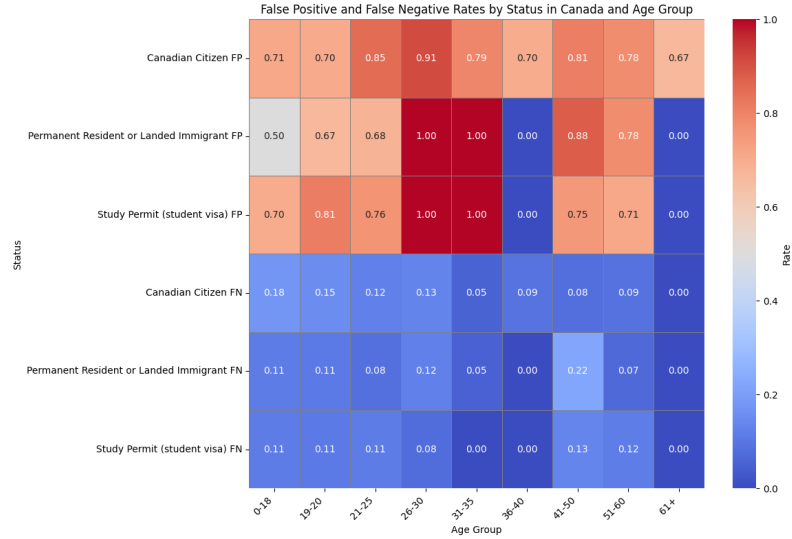


Figure 10. False Positive and False Negative Rates by Status in Canada and Age Group, Medium Risk

	Age Group	Domestic Accuracy	International Accuracy	Difference
High Risk	0	0.8051	0.7301	0.0750
	19	0.8079	0.7867	0.0212
	21	0.8423	0.7191	0.1232
	26	0.8389	0.8169	0.0220
	31	0.7674	0.6207	0.1468
	36	0.6512	0.6000	0.0512
	41	0.7660	0.5714	0.1945
	51	0.7647	NaN	NaN
	61	1.0000	NaN	NaN
Medium Risk	0	0.6956	0.8325	-0.1369
	19	0.7125	0.7975	-0.0850
	21	0.7207	0.8263	-0.1056
	26	0.7389	0.7917	-0.0528
	31	0.8267	0.8039	0.0228
	36	0.7000	0.8000	-0.1000
	41	0.7718	0.7273	0.0445
	51	0.7353	NaN	NaN
	61	0.7143	NaN	NaN
Low Risk	0	0.9059	0.9675	-0.0616
	19	0.8974	0.9679	-0.0705
	21	0.9184	0.9865	-0.0681
	26	0.9415	0.9844	-0.0429
	31	0.9141	0.9886	-0.0745
	36	0.9329	0.9784	-0.0456
	41	0.9215	0.9496	-0.0281
	51	0.9420	0.9167	0.0254
	61	1.0000	NaN	NaN
	66	1.0000	NaN	NaN

Table 12. Accuracy by Risk Level, Age Group, and Difference Between Domestic and International Models

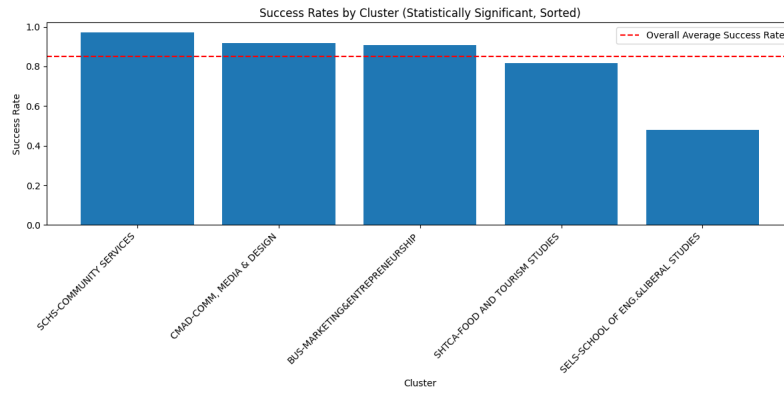


Figure 11. Cluster Success Rates International

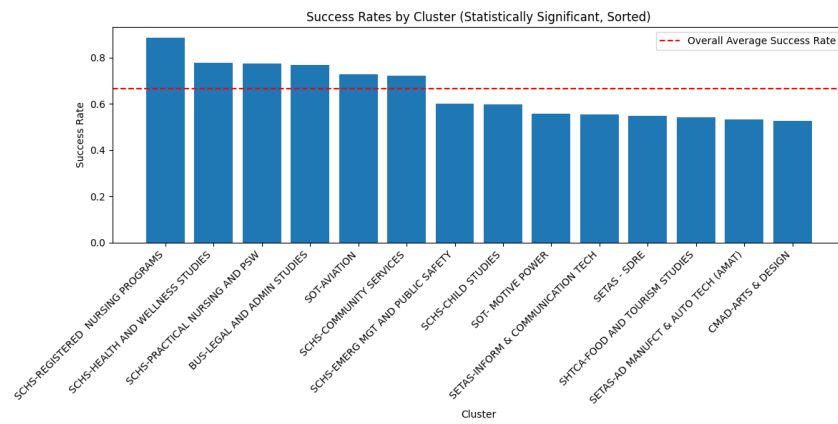


Figure 12. Cluster Success Rate Domestic